

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	BERT Emotion Prediction	Predicts primary and secondary learning emotion with confidence score from the student's text	Backend + Hugging Face Transformers	Done	High
2	BiLSTM Mixed Emotion Detection	Predicts mixed emotions and confidence scores using a Multi-label BiLSTM model	Deep Learning Engine (TensorFlow/Keras)	Done	High
3	Model Comparison View	Displays BERT vs BiLSTM predictions side by side for comparison	Frontend UI	Done	Medium
4	Subject Selection	Allows students to select the academic subject before analysis	Frontend UI	Done	Medium
5	Gemini AI Guidance Generation	Generates topic explanation, study tips, practical example, and motivation	AI Engine (Google Gemini 2.5 Flash)	Done	High
6	Personalized Study Plan	Recommends a study plan based on the detected emotion and subject	Backend Logic	Done	Medium
7	Session & CSV History	Stores and retrieves each interaction for session and long-term history	Python (Pandas, CSV)	Done	Medium
8	Analytics Dashboard	Displays emotion, subject, and summary analytics with interactive charts	Frontend (Plotly)	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Subject Selection, Model Comparison View, Analytics Dashboard
2	AI / Deep Learning Models	2	BERT Emotion Prediction, BiLSTM Mixed Emotion Detection
3	Backend / Logic	2	Personalized Study Plan, Session & CSV History
4	Generative AI / Integration	1	Gemini AI Guidance Generation
5	Security / Authentication	0	Not Used (current version has no login)